RETAILMART V3 · SQL

Self Join & UNION — Quick Quiz

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STORY

Answer Before Looking!

Test your SELF JOIN and UNION knowledge.

Quiz 1: SELF JOIN Why Two Aliases?

EASY

In a SELF JOIN, why do you need TWO different aliases for the same table?

ANSWER: Without two aliases, you can't refer to two different rows of the same table separately. Aliases e1 and e2 give you 'view 1' and 'view 2' of the table — even though it's physically one table, you treat it as two and compare rows from one alias to the other.

Quiz 2: UNION vs UNION ALL Row Count

MEDIUM

Query A returns 1000 rows. Query B returns 500 rows. 200 rows are common. UNION returns how many rows? UNION ALL?

ANSWER: UNION = 1300 rows (1000 + 500 - 200 common). UNION ALL = 1500 rows (1000 + 500, duplicates kept). UNION removes the 200 duplicates; UNION ALL preserves them.

Quiz 3: Column Mismatch

MEDIUM

ANSWER: ERROR! UNION requires the SAME number of columns in both SELECTs. Query A has 3 columns; Query B has 2. PostgreSQL throws: 'each UNION query must have the same number of columns'. Fix: align the column counts.

Question

✅ ANSWER

```
SELECT name, age, city FROM table_a
UNION
SELECT name, city FROM table_b;
```

Quiz 4: SELF JOIN Without Dedup

HARD

ANSWER: Three problems: (1) Each employee pairs with themselves: (Alice, Alice). (2) Pair (Alice, Bob) AND (Bob, Alice) both appear. (3) Row count is N-squared instead of N(N-1)/2 — run it: store 1 gives 70 rows including self-pairs. Fix: add AND a.employee_id < b.employee_id to keep each pair once, no self-pairs.

Question

✅ ANSWER

```
SELECT a.first_name, b.first_name
FROM stores.employees a
JOIN stores.employees b
  ON a.dept_id = b.dept_id
WHERE a.store_id = 1 AND b.store_id = 1;
-- (no a.employee_id < b.employee_id condition)
```

Quiz 5: UNION Column Order Matters?

CRAZY

Does the ORDER of columns inside each SELECT in a UNION matter?

ANSWER: YES, critically. UNION matches columns BY POSITION, not by name. SELECT name, age UNION SELECT email, salary will compile (both have 2 columns) but produces nonsense: name+email in column 1, age+salary in column 2. Always double-check that position N means the same thing in both queries.